Qashflow

Option Pricing with Quantum Machine Learning

Track: Finance / QML | Partner: Quandela

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Abstract

We explored quantum machine learning for swaption pricing across nine model configurations, from a classical LSTM up to a fully redesigned hybrid photonic-quantum pipeline. Our best model achieves an R2 of 0.998 on held-out data with sub-2% relative error on most prediction days.

GitHub: github.com/zeyadahmedh/Qashflow-Q-Volution-Quandela-Track
Solution: Go Through Our Solution Website

1 Introduction

Black-Scholes is elegant and almost every assumption it makes is wrong. Classical deep learning struggles when you feed it 224 features at once. Photonic quantum circuits span exponentially large feature spaces with very few physical resources. We wanted to find out how far that advantage could go.

2 Why Quantum and Why Quandela

We used **Quantum Reservoir Computing**: a fixed quantum circuit extracts non-linear features and only a lightweight classical readout is trained on top. Quandela's photonic hardware runs at room temperature and their `MerLin` library made the PyTorch integration fast to set up.

3 What We Built

We started with a classical LSTM (single layer, hidden dim 32, 200 epochs) as a baseline. Training loss reached 0.069 but validation stalled at 0.343 and would not move. From there we built a family of hybrid quantum-classical models, each using a 6-mode photonic quantum reservoir as a shared front end with different classical backbones: LSTM, GRU, Transformer, and a TCN. The TCN performed best so we swept it across 2, 3, 4, and 5 photons. Every quantum model beat the baseline. But the gains were still modest, because we were still feeding raw normalised features directly into the circuit. So we went back and redesigned the full data flow: compress first with PCA, encode into photon phase angles, run an ensemble of independent quantum reservoirs, add a photonic time-delay reservoir for temporal memory, and let a Ridge regressor close the loop.

Model	Modes	Photons	MSE	MAE	R2
Classical LSTM	—	—	0.34264	—	—
Hybrid Q-LSTM	6	2	0.20765	0.36580	—
Hybrid Q-Transformer	6	2	0.15485	0.32732	—
Hybrid Q-GRU	6	2	0.12054	0.26349	—
Hybrid Q-TCN	6	2	0.07043	0.20995	—
Hybrid Q-TCN	6	3	0.09997	0.24012	—
Hybrid Q-TCN	6	4	0.07775	0.22061	—
Hybrid Q-TCN	6	5	0.07482	0.21329	—
HPQRC	**8**	**3**	**1.93e-05**	**0.00389**	**0.998**

Table 1: All models compared. HPQRC uses the real price scale; others use normalised values.

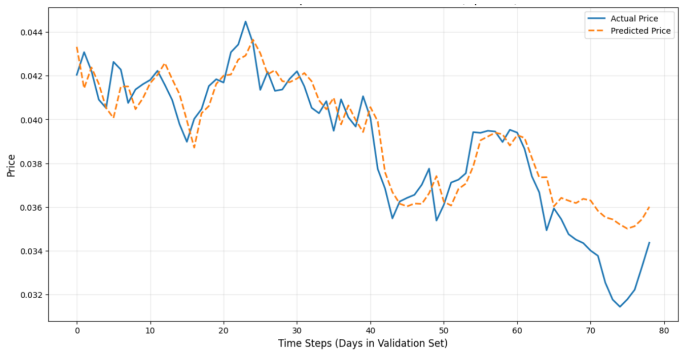

Figure 1: Classical LSTM: actual vs. predicted on the validation set.

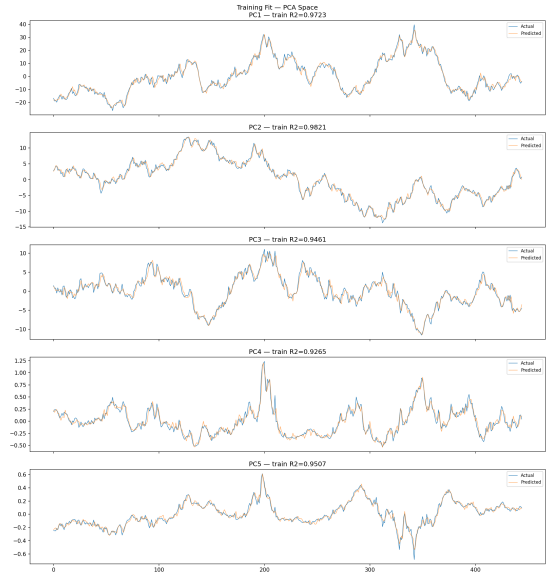

Figure 2: HPQRC fit in PCA space. R2: 0.97, 0.98, 0.95, 0.93, 0.95.

4 Our Best Model: HPQRC

Inspired by *Hybrid Photonic-Quantum Reservoir Computing for Time-Series Prediction* by Oishik Kar and Aswath Babu H. (IIIT Dharwad) [1], we asked a harder question: what if the whole pipeline was designed around what quantum hardware is actually good at, rather than just swapping in a quantum layer?

Instead of encoding 224 raw features into a circuit, we compress the swaption surface to 5 principal components with PCA, scale them into $[0, \pi]$, and run them through five independent quantum reservoirs in parallel, each seeded differently for diverse projections. A classical photonic time-delay reservoir adds temporal memory. Everything feeds into a Ridge regressor with $\alpha = 0.01$.

$$\text{prices}_{224} \to \text{PCA}(5) \to [0, \pi] \to \underbrace{\text{5x Quantum (8 modes, 3 ph.)} \oplus \text{Photonic Reservoir}}_{\text{139-dim features}} \to \text{Ridge} \to \text{prices}_{224}$$

Trained on 450 days, R2 across all five principal components ranged from 0.9265 to 0.9821. On held-out days 451 to 456, overall MSE was 1.93×10^{-5} and R2 was **0.998**, with relative error below 2% on four of six days. The only trainable component was a Ridge regressor.

5 Conclusion

We went from a model that would not generalise to one that prices a 224-dimensional swaption surface. The key was not the quantum layer alone but how the pipeline was designed around it: compress to meaningful structure first, diversify through an ensemble, and let the photonic time-delay reservoir carry the temporal memory. Quantum is not a magic layer, but give it the right problem and it is genuinely hard to beat.

References

[1] O. Kar and A. Babu H., "Hybrid Photonic-Quantum Reservoir Computing for Time-Series Prediction," IIIT Dharwad, Karnataka, India.